Mattia Corti

POSTDOCTORAL RESEARCHER

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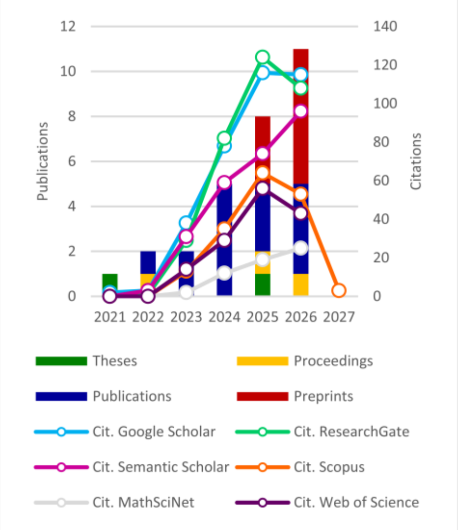
PROFILE

Postdoctoral researcher in numerical analysis. My research topic is the analysis of advanced numerical methods for PDEs with applications to the neurodegenerative diseases.

BIBLIOMETRIC INDEXES



October 14, 2026

ACADEMIC POSITIONS

Research Fellows Representative at the Department of Mathematics

Politecnico di Milano — **Jan 2026 – Present**
Department of Mathematics

Postdoctoral Research Fellow

Politecnico di Milano — **Feb 2026 – Present**
MOX Laboratory – Department of Mathematics

Position founded by the ERC Synergy Grant - NEMESIS

Principal Investigator: Prof. P.F. Antonietti

PAST ACADEMIC POSITIONS

Postdoctoral Research Fellow

Politecnico di Milano — **Mar 2025 – Feb 2026**
MOX Laboratory – Department of Mathematics

Position founded by the ERC Synergy Grant - NEMESIS

Principal Investigator: Prof. P.F. Antonietti

Postdoctoral Research Fellow

University of Vienna — **Aug 2025 – Dec 2025**
Faculty of Mathematics

Position founded by the SFB Grant – “Taming Complexity in Partial Differential Systems” Part 7

Principal Investigator: Prof. I. Perugia

PhD Student in Mathematical Models and Methods in Engineering (M^{3}E)

Politecnico di Milano — **Nov 2021 – Feb 2025**
MOX Laboratory – Department of Mathematics

Advisor: Prof. P.F. Antonietti

PhD Representative at the Department of Mathematics

Politecnico di Milano | **Jan 2023 – Dec 2024**
Department of Mathematics

INVITED RESEARCH VISITS

PhD in Mathematical Models and Methods in Engineering

Stanford University | **Mar 2024 – May 2024**
Living Matter Laboratory – Department of Mechanical Engineering

Invited by: Prof. E. Kuhl

EDUCATION

PhD in Mathematical Models and Methods in Engineering

Politecnico di Milano | **Nov 2021 – Feb 2025**
MOX Laboratory – Department of Mathematics

Thesis: "*Mathematical Models and Numerical Methods for Neurodegenerative Diseases*"

Advisor: Prof. P.F. Antonietti

Final Grade: Title conferred with honors

Master's Degree in Mathematical Engineering: Computational Science in Engineering

Politecnico di Milano | **Sep 2019 – Oct 2021**

Thesis: *"Effects of Atrial Fibrillation on Left Atrium Haemodynamics: A Patient-Specific Computational Fluid Dynamics Study"*

Advisor: Prof. A.M. Quarteroni **Co-advisors:** Prof. L. Dede', A. Zingaro

Final Grade: 110/110 cum laude

Bachelor's degree in Mathematical Engineering

Politecnico di Milano | **Oct 2016 – Sep 2019**

Thesis: *"A Bayesian approach to the inverse problem solution: neural networks vs. reduced basis method"*

Co-author: D. Fraulin **Advisor:** Prof. A. Manzoni **Co-advisor:** PhD. S. Pagani

Final Grade: 108/110

HONORS AND AWARDS

ECCOMAS PhD Award 2025	Winner of one of the two best PhD theses on computational methods in applied sciences and engineering of the European Community on Computational Methods in Applied Sciences.

Anile-ECMI Prize 2026	Winner (*ex-aequo*) of the Anile-ECMI Prize of Mathematics in Industry of European Consortium for Mathematics in Industry and Fraunhofer Institute for Industrial Mathematics - ITWM Kaiserslautern.
SIMAI Nominee for the ECCOMAS PhD Award 2025	National Italian nomination by the Italian Society of Applied and Industrial Mathematics (SIMAI) for the ECCOMAS Award for the two best PhD theses on computational methods in applied sciences and engineering.
GBMA-AIMETA PhD Thesis Award 2026	Winner (*ex-aequo*) of the Ph.D. Best Thesis Award 2026 of the Biomechanics Group (GBMA) of the Italian Association of Theoretical and Applied Mechanics (AIMETA).
Young Researchers Travel Grant Colibri Workshop 2025	Grant for Ph.D. students covering the cost of accommodation during the Colibri Focus Workshop on Computational Medicine 2025.
Switch2Product 2024	Winner of the XVI edition of Switch2Product innovation challenge (Politecnico di Milano) with the Startup AHEAD **Team:** P.F. Antonietti, M. Corti, V. Crepaldi, S. Dassatti, G. Di Fede, S. Pagani, A. Zanin

INVITED LECTURES, CONFERENCE TALKS, SEMINARS

Plenary Talks at Conferences

ECMI 2026	***Anile-ECMI Prize 2026*** recipient lecture at the 23rd "ECMI Conference on Industrial and Applied Mathematics" **Dates and Location:** June 29-July 3, 2026 - Kaunas (Lithuania)

Invited Talks at Workshops

NUMERA 2026	Invited talk at the workshop "*Numerical Advanced Methods for Real-World Application*" **Organizers:** M. Menci, G. Gargano, G. Settembre, and N. del Buono **Dates and Location:** June 15, 2026 - Bari (Italy)
New Trends in Mathematics and Brain Mechanics Workshop 2026	Invited talk at the workshop "*New Trends in Mathematics and Brain Mechanics Workshop 2026*" **Organizers:** B. Wohlmuth, P. Zunino and R. Ruiz-Baier **Dates and Location:** February 11-13, 2026 - Oslo (Norway)
LYNUM V	Invited talk at the fifth workshop "*Lombardy Young NUmerical analysts Meeting*" **Organizers:** T. Bevilacqua and M. Visinoni **Dates and Location:** May 12, 2023 – Milan (Italy)
POEMS 2022	Invited lightning talk at the workshop "*Polytopal Element Methods in Mathematics and Engineering 2022*" **Organizers:** M. Botti, F. Dassi, L. Mascotto, and I. Mazzieri **Dates and Location:** December 12-14, 2022 - Milan (Italy)

Invited Talks at Conferences

Forthcoming Talks

AIMETA 2026	***GBMA Best PhD Thesis Prize 2026*** recipient lecture at the 27th “AIMETA Congress 2026” **Dates and Location:** September 7-10, 2026 – Brescia (Italy)

Past Talks

WCCM-ECCOMAS 2026	Selected participant for the ***ECCOMAS PhD Olympiad 2026*** **Dates and Location:** July 19-24, 2026 - Munich (Germany)
WCCM-ECCOMAS 2026	Invited talk within the minisymposium "*Polytopal methods in mechanics: bridging mathematics and engineering*" **Organizers:** P. Antonietti, E. Artioli, L. Beirao da Veiga, D. Di Pietro, and P. Wriggers **Dates and Location:** July 19-24, 2026 - Munich (Germany)
GIMC-SIMAI Young 2026	Invited talk within the minisymposium "*Advances in neural network approximation and surrogate modeling for scientific machine learning*" **Organizers:** I. Bioli, E. Centofanti, F. Regazzoni, and G. Ziarelli **Dates and Location:** June 3-5, 2026 - Pisa (Italy)
SIMAI 2025	Invited talk within the minisymposium "*Efficient numerical methods for direct or inverse wave propagation problems*" **Organizers:** A. Aimi, G. Ciaramella, G. Di Credico, C. Guardasoni, and I. Mazzieri **Dates and Location:** September 1-5, 2025 - Trieste (Italy)
ICOSAHOM 2025	Invited talk within the minisymposium "*Accelerating High-Order CFD Simulations with Surrogate Modeling*" **Organizers:** P.C. Africa, F. Pichi, N. Tonicello, M. Girfoglio, and G. Rozza **Dates and Location:** July 13-18, 2025 - Montréal (Canada)
DD 29	Invited talk within the minisymposium "*Domain Decomposition Methods for Wave-Type Problems*" **Organizers:** T. Bucholz, G. Ciaramella, and R. Maier **Dates and Location:** June 23-27, 2025 – Milan (Italy)
Coupled Problems 2025	Invited talk within the minisymposium "*Computational Brain Multiphysics*" **Organizers:** P.F. Antonietti, I. Fumagalli, M. Kuchta, and M. Rognes **Dates and Location:** May 25-28, 2025 - Villasimius (Italy)
SIMAI YOUNG 2024	Invited talk within the minisymposium "*Advanced Numerical Methods for Coupled Problems on Complex Domains*" **Organizers:** A. Borio, F. Marcon, and M. Visinoni **Dates and Location:** July 10-12, 2024 - Naples (Italy)
ECCOMAS 2024	Invited talk within the minisymposium "*Microstructure-based Modeling of Biomechanical Systems*" **Organizers:** M. Terzano, S. Budday, and G.A. Holzapfel **Dates and Location:** June 3-7, 2024 - Lisbon (Portugal)
ENUMATH 2023	Invited talk within the minisymposium "*Advances in Polytopal Methods for Multiphysics Problems*" **Organizers:** P.F. Antonietti, A. Fumagalli, I. Fumagalli, and D. Prada **Dates and Location:** September 4-8, 2023 - Lisbon (Portugal)

SIMAI 2023 Invited talk within the minisymposium "*Mathematical Modelling in Biology*"
Organizers: C. Giverso, G. Lucci, G. Pozzi, and D. Riccobelli
Dates and Location: August 28 - September 1, 2023 - Matera (Italy)

Contributed Talks at Workshops

Soft Tissue Workshop 2025 Contributed talk at the workshop “*Soft Tissue Workshop 2025*”
Organizers: L. Dede', F. Migliavacca, A. Pandolfi, and C. Vergara
Dates and Location: June 11-13, 2025 - Milan (Italy)

Colibri 2025 Contributed talk at the workshop “*Colibri Focus Workshop on Computational Medicine 2025*”
Organizers: F. Caforio, A. Goritschnig, T. Grandits, G. Haase, K. L. Kapper, and E. Karabelas
Dates and Location: January 30-31, 2025 - Graz (Austria)

The Numerical Brain 2024 Lightning talk at the workshop “*The Numerical Brain 2024*”
Organizers: D. Avitabile and S. Dubinkina
Dates and Location: October 21-23, 2024 - Amsterdam (The Netherlands)

Poster Presentations at Conferences and Workshops

EMS-SciML 2025 Poster Presentation at *Annual Meeting of EMS Activity Group on Scientific Machine Learning 2025*
Organizers: P.F. Antonietti, S. Pagani, F. Regazzoni, M. Verani, and P. Zunino
Dates and Location: March 24-26, 2025 - Milan (Italy)

Student Presentations at Summer Schools

10th Summer School on Biomechanics 2023 Short talk of participant within the “*10th Summer School on Biomechanics of Soft Tissues: From the Heart to the Cardiovascular System to the Brain*”
Organizers: G.A. Holzapfel and R. Ogden
Dates and Location: September 11-15, 2023 - Graz (Austria)

Dissemination Activities

2022 Responsible of the Laboratory “Calcolo scientifico: algoritmi alla scoperta del mondo” (with I. Fumagalli) Summer School “Matematica e Scienze della Vita: modelli e algoritmi della salute” – University of Bergamo
Dates and Location: September 6-8, 2022 - San Pellegrino Terme (Italy)

INTERVIEWS, PRESS AND PODCASTS

Media Coverage

2026 "How Alzheimer’s begins and spreads in the brain"
Feature article on research published in *Comput. Methods Appl. Mech. Eng.* (Corti et al., 2026)
Politecnico di Milano News [Link]

TEACHING ACTIVITIES

Teaching Assistant: Integration of ODE based and neural network models. Intensive “Passion in Action – ENHANCE” Course. Politecnico di Milano - TU Delft. **A.Y. 2025/26**
Lecturers: L. Bonaventura, A. Heinlein **Total of Hours:** 15 Hours **Total of Students:** 25

Teaching Assistant: Numerical Mathematics. BSc in Mathematical Engineering. Politecnico di Milano. **A.Y. 2025/26**
Lecturer: M. Verani **Total of Hours:** 48 Hours **Total of Students:** 203

Teaching Assistant: Numerical Mathematics. BSc in Mathematical Engineering. Politecnico di Milano. **A.Y. 2024/25**
Lecturer: M. Verani **Total of Hours:** 48 Hours **Total of Students:** 200

Teaching Assistant: Numerical Mathematics. BSc in Mathematical Engineering. Politecnico di Milano. **A.Y. 2022/23**
Lecturer: P.F. Antonietti **Total of Hours:** 48 Hours **Total of Students:** 214

Teaching Assistant: Numerical Mathematics. BSc in Mathematical Engineering. Politecnico di Milano. **A.Y. 2021/22**
Lecturer: P.F. Antonietti **Total of Hours:** 48 Hours **Total of Students:** 222

Tutor: Fundamentals of Maths and Statistics. BSc in Urban Planning. Politecnico di Milano. **A.Y. 2020/21**
Project: 150 Hours Tutoring of Students.

MENTORING ACTIVITIES

Master Theses

[MT1] S. Di Noto - *"Numerical analysis of a modified Fisher-Kolmogorov model with application to protein spreading in neurodegenerative diseases"* **22/07/2026**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** F. Bonizzoni, F. Regazzoni, M. Corti, N. De March

[MT2] N. Ciliberti - *"On the choice of the entropy function and iterative solver in structure-preserving discontinuous methods for the Fisher Kolmogorov-Petrovsky-Piskunov Equation"* **26/03/2026**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** I. Perugia, M. Corti

[MT3] B. Caon - *"Numerical validation of physics-based and reduced graph-based computational models for neurodegenerative diseases against clinical data"* **10/12/2025**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** F. Bonizzoni, M. Corti

[MT4] V. Mazzucco - *"A mathematical and computational framework to model low to high-grade brain tumor dynamics"* **10/12/2025**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti, F. Pessina **Co-Advisors:** S. Pagani, B. Bono, M. Corti

[MT5] L. Citterio - *"A p-adaptive polytopal discontinuous Galerkin method for the Fisher-Kolmogorov equation with application to neurodegenerative disorder"* **23/10/2025**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** M. Corti, C.B. Leimer Saglio

[MT6] N. De March - *"A positivity-preserving discontinuous Galerkin method on polytopal grids for a modified Fisher–Kolmogorov model"* **02/09/2025**
23/10/2025

MSc Degree in High Performance Computing (University of Luxembourg)
MSc Degree in High Performance Computing Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** F. Regazzoni, M. Corti

[MT7] S. Minelli - *"Discontinuous Galerkin discretizations of hyperelastic models with applications to brain mechanics"* **22/07/2025**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** M. Corti

[MT8] A. Mancini - *"A one-way coupled mathematical model of Alzheimer's disease and epilepsy: A multi-scale analysis of neuronal activity"* **03/04/2025**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** S. Pagani, M. Corti, C.B. Leimer Saglio

[MT9] D. Serra - *"Discontinuous Galerkin approximation of the Smoluchowski model with application to neurodegenerative diseases progression"* **16/07/2024**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** M. Corti

[MT10] G. Bonalumi - *"Discontinuous Galerkin approximation of a coupled mathematical model of misfolded protein dynamics with application to Alzheimer's disease "* **09/04/2024**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisor:** M. Corti

[MT11] G. Lorenzon - *"Neurodegenerative disease modelling: discontinuous Galerkin approach for the heterodimer model of prions dynamics"* **09/04/2024**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisor:** M. Corti

[MT12] E. Ravizza - *"A coupled PDE-ODE mathematical model on graphs for Alzheimer's disease"* **09/04/2024**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisor:** M. Corti

[MT13] V. Pederzoli - *"A mathematical model of brain atrophy in Alzheimer's disease"* **09/04/2024**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** D. Riccobelli, M. Corti

[MT14] D.F. Marino - *"Monte Carlo and multilevel Monte Carlo methods for the Fisher-Kolmogorov equation"* **09/04/2024**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** M. Corti, F. Bonizzoni

[MT15] G. Martinelli - *"Agglomeration of polytopal grids through geometrical deep learning"* **19/12/2023**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** E. Manuzzi, M. Corti

[MT16] A. Dall'Olio - *"Discontinuous Galerkin methods for the heterodimer model of prion dynamics in neurodegenerative diseases"* **05/10/2023**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** F. Bonizzoni, M. Corti

[MT17] I. De Vittori - *"Numerical modelling of the cerebrospinal fluid flow in brain ventricles"* **05/05/2023**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** M. Corti, I. Fumagalli, F. Regazzoni

[MT18] C.B. Leimer Saglio - *"An adaptive discontinuous Galerkin method for the numerical modeling of epileptic seizures"* **05/05/2023**
MSc Degree in Mathematical Engineering (Politecnico di Milano)
Advisor: P.F. Antonietti **Co-Advisors:** S. Pagani, M. Corti

Master Degree Internships

[MI1] H. Lederman - *"Numerical discretization for the Helmholtz equation"* **15/04/2023-28/07/2023**
Supervisor: S. Fliss **Host-University Supervisors:** P.F. Antonietti, F. Bonizzoni, M. Corti

Bachelor Theses

[BT1] F. Fehr, F. Gallo, and E. Marconi - *"Modelli ionici di elettrofisiologia cellulare"* **04/03/2026**
BSc Degree in Biomedical Engineering (Politecnico di Milano)
Advisor: F. Bonizzoni **Co-Advisors:** M. Corti, C.B. Leimer Saglio

PUBLICATIONS

Peer-Reviewed Journal Papers

[J1] **M. Corti*** and S. Gómez – “On the compact discontinuous Galerkin method for polytopal meshes” – *SIAM Journal on Scientific Computing*. In press (2026)

[J2] P.F. Antonietti, **M. Corti***, S. Gómez, and I. Perugia – “Structure-preserving local discontinuous Galerkin discretization of conformational conversion systems” – *Mathematics of Computation*. In press (2026)

[J3] **M. Corti***, A. Ahern, A. Goriely, E. Kuhl, and P.F. Antonietti – “A whole-brain model of amyloid beta accumulation and cerebral hypoperfusion in Alzheimer's disease” – *Computer Methods in Applied Mechanics and Engineering*. 461. 119196 (2026) [doi]

[J4] C.B. Leimer Saglio, S. Pagani, **M. Corti**, and P.F. Antonietti – “A high-order discontinuous Galerkin method for the numerical modeling of epileptic seizures” – *Computers & Mathematics with Applications.* 205. 112-131 (2026) [doi]

[J5] P.F. Antonietti, **M. Corti***, and G. Martinelli – “Polytopal mesh agglomeration via geometrical deep learning for three-dimensional heterogeneous domains” – *Mathematics and Computers in Simulation.* 241 (Part B). 335-353 (2026) [doi]

[J6] P. F. Antonietti, **M. Corti***, S. Gómez, and I. Perugia – “A structure-preserving LDG discretization of the Fisher-Kolmogorov equation for modeling neurodegenerative diseases” – *Mathematics and Computers in Simulation.* 241 (Part A). 351-366 (2026) [doi]

[J7] S. Bonetti and **M. Corti** – “Unified discontinuous Galerkin analysis of a thermo/poro-viscoelasticity model” – *Journal of Scientific Computing.* 105. 11 (2025) [doi]

[J8] V. Pederzoli, **M. Corti**, D. Riccobelli, and P.F. Antonietti – “A coupled mathematical and numerical model for protein spreading and tissue atrophy, applied to Alzheimer's disease” – *Computer Methods in Applied Mechanics and Engineering*. 444. 118118 (2025) [doi]

[J9] P.F. Antonietti, S. Bonetti, M. Botti, **M. Corti**, I. Fumagalli, and I. Mazzieri – “lymph: discontinuous poLYtopal methods for Multi-PHysics differential problems” – *ACM Transactions on Mathematical Software.* 51 (1). 3:1-3:22 (2025) [doi]

[J10] **M. Corti***, F. Bonizzoni, P.F. Antonietti, and A.M. Quarteroni – “Uncertainty quantification for Fisher-Kolmogorov equation on graphs with applications to patient-specific Alzheimer’s disease” – *ESAIM: Mathematical Modelling and Numerical Analysis.* 58 (6). 2135-2154 (2024) [doi]

[J11] **M. Corti*** – “Exploring tau protein and amyloid-beta propagation: a sensitivity analysis of mathematical models based on biological data” – *Brain Multiphysics.* 7. 100098 (2024) [doi]

[J12] P.F. Antonietti, F. Bonizzoni, **M. Corti***, and A. Dall’Olio – “Discontinuous Galerkin approximations of the heterodimer model for protein-protein interaction” – *Computer Methods in Applied Mechanics and Engineering*. 431. 117282 (2024) [doi]

[J13] **M. Corti***, F. Bonizzoni, and P.F. Antonietti – “Structure preserving polytopal discontinuous Galerkin methods for the numerical modeling of neurodegenerative diseases” – *Journal of Scientific Computing*. 100. 39 (2024) [doi]

[J14] I. Fumagalli, **M. Corti**, N. Parolini, and P.F. Antonietti – “Polytopal discontinuous Galerkin discretization of brain multiphysics flow dynamics” – *Journal of Computational Physics*. 513. 113115 (2024) [doi]

[J15] **M. Corti***, F. Bonizzoni, L. Dede’, A.M. Quarteroni, and P.F. Antonietti – “Discontinuous Galerkin methods for Fisher-Kolmogorov Equation with application to α-synuclein spreading in Parkinson's disease” – *Computer Methods in Applied Mechanics and Engineering.* 417 (Part A). 116450 (2023) [doi]

[J16] **M. Corti***, P.F. Antonietti, L. Dede’, and A.M. Quarteroni – “Numerical modelling of the brain poromechanics by high-order discontinuous Galerkin methods” – *Mathematical Models and Methods in Applied Sciences*. 33 (8). 1577-1609 (2023) [doi]

[J17] **M. Corti***, A. Zingaro, L. Dede’, and A.M. Quarteroni – “Impact of atrial fibrillation on left atrium haemodynamics: A computational fluid dynamics study” – *Computers in Biology and Medicine*. 150. 106143 (2022) [doi]

* indicates corresponding author

Submitted Papers

[S1] **M. Corti** – “A weighted polygonal discontinuous Galerkin method for hierarchically coupled reaction–diffusion systems with cubic interactions” – *ArXiv* (2026)

[S2] P.F. Antonietti, F. Bonizzoni, **M. Corti**, N. De March, S. Di Noto, and F. Regazzoni – “A stability-preserving polytopal discontinuous Galerkin method for the Fisher-Kolmogorov model with applications to neurodegenerative disease modelling” – *ArXiv* (2026)

[S3] P.F. Antonietti, **M. Corti**, C.B. Leimer Saglio, and S. Pagani – “The lymph 2.0 library: *p*-adaptive algorithms and parallel assembly strategies for polytopal DG methods” – *ArXiv* (2026)

[S4] P.F. Antonietti, **M. Corti***, and G. Orlando – “Optimized high-order IMEX-RK schemes for degenerate diffusion-reaction problems with application to travelling waves phenomena” – *ArXiv* (2026)

[S5] B. Caon, **M. Corti**, F. Bonizzoni, and P.F. Antonietti – "High-fidelity and network-based spatio-temporal mathematical models of Alzheimer's disease progression and their validation against PET-SUVR imaging data" – *ArXiv* (2026)

[S6] C.B. Leimer Saglio, **M. Corti**, S. Pagani, P.F. Antonietti – "A novel mathematical and computational framework of amyloid-beta triggered seizure dynamics in Alzheimer's disease" – *ArXiv* (2025)

[S7] A. Zanin, S. Pagani, **M. Corti**, V. Crepaldi, G. Di Fede, P.F. Antonietti, and for the Alzheimer's Disease Neuroimaging Initiative (ADNI) – "Predicting Alzheimer's disease progression from sparse multimodal data by NeuralODE models" – *BiorXiv* (2025)

Conference Proceedings

[P1] P.F. Antonietti, **M. Corti**, and G. Lorenzon – "A discontinuous Galerkin method for the three-dimensional heterodimer model with application to prion-like proteins' dynamics" – *European Congress of Mathematics Sevilla, 15–19 July, 2024.* 217-240. EMS Press. (2026) [doi]

[P2] P.F. Antonietti and **M. Corti** – "Numerical modelling of protein misfolding in neurodegenerative diseases: a computational study" – *Numerical Mathematics and Advanced Applications ENUMATH 2023, Volume 1: European Conference, September 4-8, Lisbon, Portugal.* 1-16. Lecture Notes in Computational Science and Engineering, 153. Springer Nature Switzerland (2025) [doi]

[P3] A. Zingaro, **M. Corti**, L. Dede', and A.M. Quarteroni – "Effects of atrial fibrillation on the left atrium hemodynamics" – *Proceedings of the 7th International Conference on Computational and Mathematical Biomedical Engineering - CMBE2021, Milan* (2022)

Theses

[T1] **M. Corti** - *Mathematical models and numerical methods for neurodegenerative diseases* – PhD Thesis in Mathematical Models and Methods in Engineering (2025) [url]

[T2] **M. Corti** - *Effects of atrial fibrillation on left atrium haemodynamics: a patient-specific computational fluid dynamics study* - Master Thesis in Mathematical Engineering: Computational Science in Engineering (2021)

[T3] **M. Corti** and D. Fraulin - *Un approccio Bayesiano per la soluzione di problemi inversi: reti neurali e metodi a basi ridotte a confronto* - Tesi di Laurea Triennale in Ingegneria Matematica (2019)

ORGANIZATION OF SCIENTIFIC ACTIVITIES

Organization of Conferences and Workshops

DD29 Member of local support committee of the 29th edition of the International Conference on Domain Decomposition Methods
Dates and Location: June 23-27, 2025 – Milan (Italy)

LYNUM VI Organization of the sixth edition of the Lombardy Young NUmerical analysts Meeting
Organizers: M. Corti and E. Temellini
Dates and Location: June 14, 2024 – Milan (Italy)

Organization of Minisymposia at Conferences

Coupled Problems 2027	Organization of the minisymposium "*Computational Brain Multiphysics*" within the XII International Conference on Coupled Problems in Science and Engineering **Organizers:** M. Corti, I. Fumagalli, M. Kuchta, and M. Rognes **Dates and Location:** June 27-30, 2027 – Palma de Mallorca (Spain)
SIMAI 2025	Organization of the minisymposium "*Advances in Polytopal Methods with Applications in Engineering*" within the SIMAI Congress 2025 **Organizers:** S. Bonetti, M. Botti, A. Cangiani, M. Corti, and I. Fumagalli **Dates and Location:** September 1-5, 2025 – Trieste (Italy)
ICOSAHOM 2025	Organization of the minisymposium "*Advances in Numerical Methods for Multi-Physics Problems and Applications*" within the 15th International Congress on Spectral and High Order Methods **Organizers:** S. Bonetti, F. Bonizzoni, M. Corti, F. Dessi, and I. Fumagalli **Dates and Location:** July 13-18, 2025 – Montréal (Canada)
DD 29	Organization of the minisymposium "*Next-Generation Numerical Methods for Computational Life Sciences*" within the XI International Conference on Coupled Problems in Science and Engineering **Organizers:** F. Bonizzoni, M. Corti, I. Fumagalli, and S. Pagani **Dates and Location:** June 23-27, 2025 – Milan (Italy)
Coupled Problems 2025	Organization of the minisymposium "*High-Order and Innovative Methods for Coupled Problems in Life Science and Geophysics*" within the XI International Conference on Coupled Problems in Science and Engineering **Organizers:** F. Bonizzoni, M. Corti, and S. Pagani **Dates and Location:** May 25-28, 2025 – Villasimius (Italy)

REVIEWER ACTIVITY

Peer-Reviewed Academic Journals

- Reviewer for 2 articles on *Annals of Biomedical Engineering* (2025)
- Reviewer for 2 articles on *Computers and Mathematics with Applications* (2024-**2026**)
- Reviewer for 2 articles on *Computer Methods in Applied Mechanics and Engineering* (2025-**2026**)
- Reviewer for 2 articles on *Journal of Mathematical Biology* (**2026**)
- Reviewer for 1 article on *Applied Mathematics and Computation* (2023)
- Reviewer for 1 article on *Applied Numerical Mathematics* (**2026**)
- Reviewer for 1 article on *Brain Research Bulletin* (2025)
- Reviewer for 1 article on *Communications in Applied and Industrial Mathematics* (2025)
- Reviewer for 1 article on *Communications in Computational Physics* (2025)
- Reviewer for 1 article on *Journal of Scientific Computing* (2024)
- Reviewer for 1 article on *SIAM Journal on Scientific Computing* (**2026**)

Peer-Reviewed Conference Proceedings

- Reviewer for 1 article on *Numerical Mathematics and Advanced Applications ENUMATH 2023, Volume 1: European Conference, September 4-8, Lisbon, Portugal* (2024)

Abstracting and Reviewing Online Services

- Reviewer for 1 article on *Zentralblatt MATH Open* (2025)

FUNDED RESEARCH PROJECTS

Principal Investigator

2026-2027	GNCS Research Project 2026, *Metodi poligonali di Galerkin discontinui per equazioni alle derivate parziali non lineari* (CUP E53C25002010001), INdAM, GNCS, Italy. Amount: € 1.600
2025-2026	ISCRA C project, *NeuroDG - Mathematical Models and Methods in Neurodegenerative Diseases Modeling,* CINECA, Italy. Amount: € 1.920 (=96k core hours on G100 cluster, CINECA, Italy)

PROFESSIONAL SOCIETIES

- Member of **AIMETA** (Associazione Italiana di Meccanica Teorica e Applicata) **2026-Present**
- Member of **SIMAI** (Società Italiana di Matematica Applicata e Industriale) **2023-Present**
- Member of **Indam-GNCS** (Gruppo Nazionale di Calcolo Scientifico dell'Istituto Nazionale di Alta Matematica "Francesco Severi") **2022-Present**

SOFTWARE CONTRIBUTION

`lymph` **Source:** https://bitbucket.org/lymph/lymph

MATLAB library for the discretization of PDEs based on high-order discontinuous Galerkin methods on polytopal grids (PolyDG) for spatial discretization coupled with suitable finite-difference time marching schemes. (The lymph library is also publicly available on the MathWorks website).

lymph

WEB ACADEMIC PROFILES AND METRICS

Google Scholar	**Citations**	353 (353 from 2021)
	h-index	10 (10 from 2021)
	i10-index	11 (11 from 2021)

scholar.google.com/citations?user=xhqRbvUAAAAJ&hl=it

ResearchGate	**Citations**	343
	h-index	10
	RI Score	376.6

www.researchgate.net/profile/Mattia-Corti

Scopus	**Citations**	168
	h-index	8

www.scopus.com/authid/detail.uri?authorId=57477394000

Semantic Scholar	**Citations**	263
	h-index	10
	Highly Influential Citations	10

www.semanticscholar.org/author/Mattia-Corti/2068145801

MathSciNet	**Citations**	58
	h-index	4

https://mathscinet.ams.org/mathscinet/author?authorId=1566252

Web of Science	**Citations**	142
	h-index	6

https://www.webofscience.com/wos/author/record/KDN-0229-2024